

1. $(x^a)' =$
2. $(a^x)' =$
3. $(e^x)' =$
4. $(\log_a x)' =$
5. $(\ln |x|)' =$
6. $(\sin x)' =$
7. $(\cos x)' =$
8. $(\tan x)' =$
9. $(\cot x)' =$
10. $(\sec x)' =$
11. $(\csc x)' =$
12. $(\ln |\cos x|)' =$
13. $(\ln |\sin x|)' =$
14. $(\ln |\sec x + \tan x|)' =$
15. $(\ln |\csc x - \cot x|)' =$
16. $(\ln |\csc x + \cot x|)' =$
17. $(\arcsin x)' =$
18. $(\arccos x)' =$
19. $(\arctan x)' =$
20. $(\operatorname{arccot} x)' =$
21. $[\ln(x + \sqrt{x^2 + 1})]' =$
22. $[\ln(x + \sqrt{x^2 - 1})]' =$
23. $[\ln(x + \sqrt{x^2 + a^2})]' =$
24. $(\frac{1}{2} \arcsin x + \frac{x}{2} \sqrt{1 - x^2})' =$

25. $[\ln(x + \sqrt{x^2 - a^2})]' =$
26. $(x \ln x)' =$
27. $(\frac{1}{x} \ln x)' =$
28. $(e^{ax+b})^{(n)} =$
29. $[\sin(ax + b)]^{(n)} =$
30. $[\cos(ax + b)]^{(n)} =$
31. $[\ln(ax + b)]^{(n)} =$
32. $\left(\frac{1}{ax + b}\right)^{(n)} =$
33. $(\frac{1}{1 + x})^{(n)} =$
34. $\int x^k \, dx =$
35. $\int \frac{1}{x} \, dx =$
36. $\int e^x \, dx =$
37. $\int a^x \, dx =$
38. $\int \sin x \, dx =$
39. $\int \cos x \, dx =$
40. $\int \tan x \, dx =$
41. $\int \cot x \, dx =$
42. $\int \frac{1}{\cos x} \, dx =$

43. $\int \frac{1}{\sin x} \, dx =$
44. $\int \frac{1}{\cos^2 x} \, dx =$
45. $\int \frac{1}{\sin^2 x} \, dx =$
46. $\int \sec x \tan x \, dx =$
47. $\int \csc x \cot x \, dx =$
48. $\int \frac{1}{1 + x^2} \, dx =$
49. $\int \frac{1}{a^2 + x^2} \, dx =$
50. $\int \frac{1}{\sqrt{1 - x^2}} \, dx =$
51. $\int \frac{1}{\sqrt{a^2 - x^2}} \, dx =$
52. $\int \sqrt{1 - x^2} \, dx =$
53. $\int \sqrt{a^2 - x^2} \, dx =$
54. $\int \frac{1}{\sqrt{x^2 + 1}} \, dx =$
55. $\int \frac{1}{\sqrt{x^2 + a^2}} \, dx =$
56. $\int \frac{1}{\sqrt{x^2 - 1}} \, dx =$
57. $\int \frac{1}{\sqrt{x^2 - a^2}} \, dx =$
58. $\int \sqrt{1 + x^2} \, dx =$

$$59. \int \frac{1}{x^2 - a^2} dx =$$

$$61. \int \sin^2 x dx =$$

$$63. \int \tan^2 x dx =$$

$$60. \int \frac{1}{a^2 - x^2} dx =$$

$$62. \int \cos^2 x dx =$$

$$64. \int \cot^2 x dx =$$

$$65. \int_0^{\frac{\pi}{2}} \sin^n x dx =$$

$$69. \int_0^{2\pi} \sin^n x dx =$$

$$66. \int_0^{\frac{\pi}{2}} \cos^n x dx =$$

$$70. \int_0^{2\pi} \cos^n x dx =$$

$$67. \int_0^{\pi} \sin^n x dx =$$

$$71. \int e^{ax} \cos bx dx =$$

$$68. \int_0^{\pi} \cos^n x dx =$$

$$72. \int e^{ax} \sin bx dx =$$

$$1. \sin\left(\frac{\pi}{2} \pm \alpha\right) =$$

$$6. \cos(\alpha \pm \beta) =$$

$$11. \cos 2\alpha =$$

$$2. \sin(\pi \pm \alpha) =$$

$$7. \tan(\alpha \pm \beta) =$$

$$12. \sin 3\alpha =$$

$$3. \cos\left(\frac{\pi}{2} \pm \alpha\right) =$$

$$8. \tan\left(\frac{\pi}{4} - \alpha\right) =$$

$$13. \cos 3\alpha =$$

$$4. \cos(\pi \pm \alpha) =$$

$$9. \cot(\alpha \pm \beta) =$$

$$14. \sin^2 \frac{\alpha}{2} =$$

$$5. \sin(\alpha \pm \beta) =$$

$$10. \sin 2\alpha =$$

$$15. \cos^2 \frac{\alpha}{2} =$$

$$1. e^x =$$

$$2. \sin x =$$

$$3. \cos x =$$

$$4. \arctan x =$$

$$5. \frac{1}{1-x} =$$

$$6. \frac{1}{1+x} =$$

$$7. \ln(1+x) =$$

$$8. (1+x)^a =$$

$$9. \tan x =$$

$$\arcsin x =$$

1. 七种未定式：

2. 等差数列通项公式：

3. 等差数列前 n 项和：

4. 等比数列通项公式：

5. 等比数列前 n 项和：

6. $2f(x) \cdot f'(x) =$

7. $[f'(x)]^2 + f(x)f''(x) =$

8. $[f'(x) + f(x)\varphi'(x)]e^{\varphi(x)} =$

9. $f'(x) + f(x) =$

10. $f'(x) - f(x) =$

11. $f'(x) + kf(x) =$

12. 曲率公式：

13. 曲率半径公式：

14. 牛顿-莱布尼茨公式：

15. 区间再现公式：

16. $u''v + 2u'v' + uv'' =$

17. $\frac{xf'(x) - f(x)}{x^2} =$

18. $\frac{f''(x)f(x) - [f'(x)]^2}{f^2(x)} =$

19. $\frac{f''(x)f(x) - [f'(x)]^2}{f^2(x)} =$

20. $\frac{f'(x)}{f(x)} =$

1. 有界与最值定理：

2. 介值定理：

3. 平均值定理：

4. 零点定理：

5. 费马定理：

6. 导数零点定理：

7. 罗尔定理：

8. 拉格朗日中值定理：

9. 柯西中值定理：

1. 三角不等式

2. 反三角不等式

3. 基本不等式

4. 三元均值不等式

5. $0 < a < x < b, 0 < c < y < d$

6. $e^x \geq 1 + x$

7. $\ln x \leq x - 1 \ (x > 0)$

8. $\frac{1}{1+x} \leq \ln\left(1 + \frac{1}{x}\right) \leq \frac{1}{x}, (x > 0)$

9. $\frac{x}{1+x} \leq \ln(1+x) \leq x, (x > 0)$

10. $0 < x < \frac{\pi}{2}$

11. $\frac{1}{n^2+n} \leq \frac{1}{n^2+k} \leq \frac{1}{n^2+1}$

12. $n \leq \sqrt{n^2+k} \leq \sqrt{n^2+n}$

13. $x > 0, \sin x \leq x$

14. $0 < x < \frac{\pi}{4}$

15. $x > 0, \arctan x \leq x \leq \arcsin x$